

SUPPLEMENTARY MATERIALS: SUPPLEMENTARY MATERIALS: A JOINT-DISTRIBUTION ROUTE TO FAIR REPRESENTATIONS WITH CONTINUOUS SENSITIVE ATTRIBUTES*

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SM1. Proofs. This section collects the proofs of the results stated in the main paper, in the order in which the results appear there. The technical lemma used in the proof of Theorem 3.6 (Lemma SM2.1) is recorded in Section SM2 below.

SM1.1. Proof of Proposition 3.1: conditions (1)–(4).

Proof. We show $(i) \Rightarrow (ii) \Rightarrow (iii) \Rightarrow (iv) \Rightarrow (i)$.

$(i) \Rightarrow (ii)$. Under characteristic k_Z, k_S , the product kernel $k_Z \otimes k_S$ is $\mathcal{I}$ -characteristic (Szabó and Sriperumbudur 2018, Theorem 3, $M = 2$), so $\mu_{Z,S} = \mu_Z \otimes \mu_S$ implies $P_{Z,S} = P_Z \otimes P_S$.

$(ii) \Rightarrow (iii)$. On the Polish product space $\mathcal{Z} \times \mathcal{S}$, the disintegration theorem (Kallenberg, 2002, Thm. 5.4) yields a regular conditional probability $\{P_{Z|S=s}\}_{s \in \mathcal{S}}$ such that for every Borel $A \subseteq \mathcal{Z}, B \subseteq \mathcal{S}$,

$$P_{Z,S}(A \times B) = \int_B P_{Z|S=s}(A) dP_S(s).$$

Under (ii) , the same Borel set $A \times B$ also satisfies $P_{Z,S}(A \times B) = P_Z(A) P_S(B) = \int_B P_Z(A) dP_S(s)$. Comparing the two expressions, $\int_B [P_{Z|S=s}(A) - P_Z(A)] dP_S(s) = 0$ for every Borel B , so $P_{Z|S=s}(A) = P_Z(A)$ for P_S -a.e. s . Choose a countable algebra $\mathcal{A}$ generating the Borel σ -algebra of $\mathcal{Z}$ (which exists since $\mathcal{Z}$ is Polish); the exceptional P_S -null sets, taken over the countable family $\mathcal{A}$, have countable union of P_S -measure zero. Hence $P_{Z|S=s}|_{\mathcal{A}} = P_Z|_{\mathcal{A}}$ for P_S -a.e. s , and by the π - λ theorem $P_{Z|S=s} = P_Z$ on the full Borel σ -algebra for P_S -a.e. s . Equivalence of “some” and “every” regular conditional version follows from the P_S -a.e. uniqueness clause of the disintegration theorem.

$(iii) \Rightarrow (iv)$. For every bounded measurable f and P_S -a.e. s ,

$$\mathbb{E}[f(Z) | S = s] = \int f(z) dP_{Z|S=s}(z) = \int f(z) dP_Z(z) = \mathbb{E}[f(Z)],$$

where the second equality uses (iii) .

$(iv) \Rightarrow (i)$. Indicator functions of Borel sets are bounded and measurable, so applying (iv) with $f = \mathbf{1}_A$ gives $P_{Z|S=s}(A) = P_Z(A)$ for P_S -a.e. s , for every Borel $A \subseteq \mathcal{Z}$. Take a countable algebra $\mathcal{A}$ generating the Borel σ -algebra of the Polish space $\mathcal{Z}$; the exceptional P_S -null sets over the countable family $\mathcal{A}$ have null union, so $P_{Z|S=s}|_{\mathcal{A}} = P_Z|_{\mathcal{A}}$ for P_S -a.e. s , and the π - λ theorem gives $P_{Z|S=s} = P_Z$ for P_S -a.e. s . Hence $Z \perp S$, so the joint mean embedding factorizes, $\mu_{Z,S} = \mu_Z \otimes \mu_S$, and $\text{HSIC}(Z, S) = \|\mu_{Z,S} - \mu_Z \otimes \mu_S\|_{\mathcal{F}_{Z \otimes S}}^2 = 0$. $\square$

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SM1.2. Proof of Proposition 3.1: equivalence with EIPM.

Proof. ($\Rightarrow$) If $\text{HSIC}(Z, S) = 0$, then $\mu_{Z,S} = \mu_Z \otimes \mu_S$ in $\mathcal{F}_{Z \otimes S}$. Because k_Z and k_S are characteristic, the product kernel $k_Z \otimes k_S$ is $\mathcal{I}$ -characteristic, i.e., its joint mean embedding distinguishes $P_{Z,S}$ from $P_Z \otimes P_S$ (Theorem 3 of Szabó and Sriperumbudur, 2018, taking $M = 2$). Hence $\mu_{Z,S} = \mu_Z \otimes \mu_S$ implies $P_{Z,S} = P_Z \otimes P_S$, i.e., $Z \perp S$. Hence $P_{Z|S=s} = P_Z$ for P_S -almost every s , so $\text{MMD}(P_{Z|S=s}, P_Z) = 0$ a.s., and $\text{EIPM}_{\mathcal{V}}(Z; S) = \mathbb{E}_S[\text{MMD}(P_{Z|S=s}, P_Z)] = 0$.

($\Leftarrow$) Suppose $\text{EIPM}_{\mathcal{V}}(Z; S) = 0$. Since $\text{MMD}(P_{Z|S=s}, P_Z) \geq 0$, this implies $\text{MMD}(P_{Z|S=s}, P_Z) = 0$ for P_S -almost every s . Because k_Z is characteristic, MMD is a metric on probability distributions, so $P_{Z|S=s} = P_Z$ a.s., i.e., $Z \perp S$. Independence then implies that the joint mean embedding factorizes, $\mu_{Z,S} = \mu_Z \otimes \mu_S$, so $\text{HSIC}(Z, S) = \|\mu_{Z,S} - \mu_Z \otimes \mu_S\|_{\mathcal{F}_{Z \otimes S}}^2 = 0$. $\square$

SM1.3. Proof of Proposition 3.1: downstream-head (GDP) form. By the equivalence of conditions (1) and (3) of Proposition 3.1, $\text{HSIC}(Z, S) = 0$ implies $P_{Z|S=s} = P_Z$ for P_S -almost every s , hence $\mathbb{E}[f(Z) \mid S = s] = \mathbb{E}[f(Z)]$ for P_S -almost every s and every measurable $f : \mathcal{Z} \rightarrow \mathbb{R}$ with $\mathbb{E}|f(Z)| < \infty$. Therefore $\Delta_{\text{GDP}}(f) = \mathbb{E}_S[\mathbb{E}[f(Z) \mid S] - \mathbb{E}[f(Z)]] = 0$.

SM1.4. Proof of Theorem 3.3 (Disintegration of the joint-vs-product IPM). Because $\mathcal{Z}$ and $\mathcal{S}$ are standard Borel, the disintegration theorem (Kallenberg, 2002, Theorem 5.4) gives $P_{Z,S}(dz ds) = P_{Z|S=s}(dz) P_S(ds)$, and by construction $(P_Z \otimes P_S)(dz ds) = P_Z(dz) P_S(ds)$. Fix $f \in \mathcal{F}$ with $\|f\|_{\infty} \leq b < \infty$. Both $P_{Z|S=s}(dz) P_S(ds)$ and $P_Z(dz) P_S(ds)$ are probability measures and $|f| \leq b$, so each iterated integral below is absolutely convergent and the Fubini–Tonelli theorem applies:

$$\mathbb{E}_{P_{Z,S}}[f] - \mathbb{E}_{P_Z \otimes P_S}[f] = \int_S \left(\int_{\mathcal{Z}} f(z, s) P_{Z|S=s}(dz) - \int_{\mathcal{Z}} f(z, s) P_Z(dz) \right) P_S(ds).$$

The inner difference is $\int_{\mathcal{Z}} f(z, s) d(P_{Z|S=s} - P_Z)(z)$. Taking absolute values and the supremum over $f \in \mathcal{F}$ yields the identity.

SM1.5. Proof of Corollary 3.5 (Recovery of the conditional-integral IPM). By Theorem 3.3,

$$\text{IPM}_{\mathcal{F}}(P_{Z,S}, P_Z \otimes P_S) = \sup_{f \in \mathcal{F}} \left| \mathbb{E}_S \int_{\mathcal{Z}} f(z, S) d(P_{Z|S} - P_Z)(z) \right|.$$

Upper bound. Fix $f \in \mathcal{F}$. By Definition 3.4(2), $f(\cdot, s) \in \mathcal{V}$ for P_S -a.e. s , so for those s

$$\left| \int_{\mathcal{Z}} f(z, s) d(P_{Z|S=s} - P_Z)(z) \right| \leq \sup_{g \in \mathcal{V}} \left| \int_{\mathcal{Z}} g d(P_{Z|S=s} - P_Z) \right| = d_{\mathcal{V}}(P_{Z|S=s}, P_Z).$$

By the triangle inequality for the P_S -integral, $|\mathbb{E}_S \int f(z, S) d(P_{Z|S} - P_Z)| \leq \mathbb{E}_S[d_{\mathcal{V}}(P_{Z|S}, P_Z)]$; taking the supremum over $f \in \mathcal{F}$ gives $\text{IPM}_{\mathcal{F}} \leq \mathbb{E}_S[d_{\mathcal{V}}(P_{Z|S}, P_Z)]$.

Lower bound. Fix $\varepsilon > 0$. Since $\mathcal{V}$ is symmetric ($g \in \mathcal{V} \Rightarrow -g \in \mathcal{V}$), $d_{\mathcal{V}}(P_{Z|S=s}, P_Z) = \sup_{g \in \mathcal{V}} (\int g dP_{Z|S=s} - \int g dP_Z)$, the absolute value being attained because $-g \in \mathcal{V}$. By the selector assumption of Corollary 3.5 there is a measurable map $s \mapsto g_s^{\varepsilon} \in \mathcal{V}$ with $\int g_s^{\varepsilon} d(P_{Z|S=s} - P_Z) \geq d_{\mathcal{V}}(P_{Z|S=s}, P_Z) - \varepsilon$ for P_S -a.e. s . By Definition 3.4(1), $f_{\star}(z, s) := g_s^{\varepsilon}(z)$ lies in $\mathcal{F}$, and since the integrand is nonnegative up to ε ,

$$\text{IPM}_{\mathcal{F}} \geq \left| \mathbb{E}_S \int g_s^{\varepsilon} d(P_{Z|S} - P_Z) \right| \geq \mathbb{E}_S[d_{\mathcal{V}}(P_{Z|S}, P_Z)] - \varepsilon.$$

Letting $\varepsilon \downarrow 0$ gives $\text{IPM}_{\mathcal{F}} \geq \mathbb{E}_S[d_{\mathcal{V}}(P_{Z|S}, P_Z)]$. Combining the two bounds and recalling $\mathcal{I}_{d_{\mathcal{V}}}(Z; S) = \mathbb{E}_S[d_{\mathcal{V}}(P_{Z|S}, P_Z)]$ from the preliminaries yields the identity.

SM1.6. Proof of Theorem 3.6 (HSIC and Expected MMD).

Proof. Step 1: Decomposition. By definition of the joint mean embedding and the tower property of conditional expectation,

$$\mu_{Z,S} = \mathbb{E}_{Z,S}[k_Z(Z, \cdot) \otimes k_S(S, \cdot)] = \mathbb{E}_S[\mu_{Z|S} \otimes k_S(S, \cdot)],$$

and by the definition $\mu_S = \mathbb{E}_{S'}[k_S(S', \cdot)]$ and linearity of the tensor product (with μ_Z constant in S'), $\mu_Z \otimes \mu_S = \mathbb{E}_{S'}[\mu_Z \otimes k_S(S', \cdot)]$. Subtracting and using linearity of the Bochner integral (Lemma SM2.1),

$$(SM1.1) \quad \mu_{Z,S} - \mu_Z \otimes \mu_S = \mathbb{E}_S[(\mu_{Z|S} - \mu_Z) \otimes k_S(S, \cdot)].$$

Squaring the norm and writing it as an inner product gives, with $D_S := (\mu_{Z|S} - \mu_Z) \otimes k_S(S, \cdot)$,

$$\begin{aligned} \text{HSIC}(Z, S) &= \langle \mathbb{E}_S[D_S], \mathbb{E}_{S'}[D_{S'}] \rangle_{\mathcal{F}_{Z \otimes S}} \\ &= \mathbb{E}_{S,S'}[\langle D_S, D_{S'} \rangle_{\mathcal{F}_{Z \otimes S}}], \end{aligned}$$

where the exchange of expectation and inner product is justified by Lemma SM2.1 together with the bound $\|(\mu_{Z|S} - \mu_Z) \otimes k_S(S, \cdot)\|_{\mathcal{F}_{Z \otimes S}} \leq 2\sqrt{\kappa_Z \kappa_S}$. By the tensor-product inner product identity $\langle a \otimes b, a' \otimes b' \rangle = \langle a, a' \rangle \langle b, b' \rangle$ and the reproducing property $\langle k_S(S, \cdot), k_S(S', \cdot) \rangle_{\mathcal{F}_S} = k_S(S, S')$,

$$\text{HSIC}(Z, S) = \mathbb{E}_{S,S'}[\langle \mu_{Z|S} - \mu_Z, \mu_{Z|S'} - \mu_Z \rangle_{\mathcal{F}_Z} \cdot k_S(S, S')],$$

which establishes (3.1).

Step 2: Squared expected MMD bound. Starting from (SM1.1), apply the triangle inequality (Jensen's inequality for the convex norm):

$$\begin{aligned} \text{HSIC}(Z, S)^{1/2} &= \|\mathbb{E}_S[(\mu_{Z|S} - \mu_Z) \otimes k_S(S, \cdot)]\|_{\mathcal{F}_{Z \otimes S}} \\ &\leq \mathbb{E}_S[\|(\mu_{Z|S} - \mu_Z) \otimes k_S(S, \cdot)\|_{\mathcal{F}_{Z \otimes S}}]. \end{aligned}$$

The tensor-product norm factorizes: $\|a \otimes b\|_{\mathcal{F}_{Z \otimes S}} = \|a\|_{\mathcal{F}_Z} \|b\|_{\mathcal{F}_S}$, and $\|k_S(S, \cdot)\|_{\mathcal{F}_S} = \sqrt{k_S(S, S)} \leq \sqrt{\kappa_S}$. Therefore

$$\text{HSIC}(Z, S)^{1/2} \leq \mathbb{E}_S[\|\mu_{Z|S} - \mu_Z\|_{\mathcal{F}_Z} \sqrt{k_S(S, S)}] \leq \sqrt{\kappa_S} \cdot \mathbb{E}_S[\text{MMD}_{k_Z}(P_{Z|S}, P_Z)].$$

Squaring both sides yields (3.2). □

SM1.7. Proof of Theorem 3.8 (Smoothed conditional-distribution control by HSIC).

Proof. Since k_Z is bounded, $\|\mu_{Z|S=S} - \mu_Z\|_{\mathcal{F}_Z} \leq \sqrt{\kappa_Z}$ and $\|\mu_Z\|_{\mathcal{F}_Z} \leq \sqrt{\kappa_Z}$, so $\Delta \in L^2(P_S; \mathcal{F}_Z)$. The joint mean embedding satisfies

$$\mu_{Z,S} - \mu_Z \otimes \mu_S = \mathbb{E}_S[(\mu_{Z|S=S} - \mu_Z) \otimes k_S(S, \cdot)] = \mathbb{E}_S[\Delta(S) \otimes k_S(S, \cdot)].$$

Taking the squared norm in $\mathcal{F}_{Z \otimes S}$ gives the first identity.

For $g \in \mathcal{F}_S$, the partial contraction of the tensor against g over the $\mathcal{F}_S$ factor gives an element of $\mathcal{F}_Z$:

$$(\text{id}_{\mathcal{F}_Z} \otimes \langle \cdot, g \rangle_{\mathcal{F}_S})(\mathbb{E}_S[\Delta(S) \otimes k_S(S, \cdot)]) = \mathbb{E}_S[g(S) \Delta(S)].$$

Hence

$$\|\mathbb{E}_S[g(S) \Delta(S)]\|_{\mathcal{F}_Z} \leq \|g\|_{\mathcal{F}_S} \|\mathbb{E}_S[\Delta(S) \otimes k_S(S, \cdot)]\|_{\mathcal{F}_Z \otimes \mathcal{F}_S},$$

which, with the first identity, proves (3.3). Two equivalent reformulations follow. Writing $m_f(s) := \mathbb{E}[f(Z) \mid S = s] - \mathbb{E}[f(Z)] = \langle f, \Delta(s) \rangle_{\mathcal{F}_Z}$ for $f \in \mathcal{F}_Z$, the same contraction argument applied to the $\mathcal{F}_Z$ -coordinate gives the fixed-head form

$$(SM1.2) \quad \|\mathbb{E}_S[m_f(S) k_S(S, \cdot)]\|_{\mathcal{F}_S}^2 \leq \|f\|_{\mathcal{F}_Z}^2 \text{HSIC}(Z, S).$$

Finally, since $\mathbb{E}_S[g(S) m_f(S)] = \langle f, \mathbb{E}_S[g(S) \Delta(S)] \rangle_{\mathcal{F}_Z}$, Cauchy–Schwarz together with (3.3) gives the bilinear form

$$(SM1.3) \quad |\mathbb{E}_S[g(S) m_f(S)]|^2 \leq \|f\|_{\mathcal{F}_Z}^2 \|g\|_{\mathcal{F}_S}^2 \text{HSIC}(Z, S)$$

for all $f \in \mathcal{F}_Z$ and $g \in \mathcal{F}_S$. $\square$

SM1.8. Proof of Corollary 3.9 (Localized sensitive-value contrasts).

Proof. Apply Inequality (3.3) with $g = k_S(\cdot, s_0) \in \mathcal{F}_S$: by the reproducing property $g(S) = k_S(S, s_0)$ and $\|g\|_{\mathcal{F}_S}^2 = \langle k_S(\cdot, s_0), k_S(\cdot, s_0) \rangle_{\mathcal{F}_S} = k_S(s_0, s_0)$, so $\|\mathbb{E}_S[k_S(S, s_0) \Delta(S)]\|_{\mathcal{F}_Z}^2 \leq k_S(s_0, s_0) \text{HSIC}(Z, S)$, which is the single-point bound since $\Delta(S) = \mu_{Z|S} - \mu_Z$. The two-point form follows by the same argument with $g = k_S(\cdot, s_0) - k_S(\cdot, s_1) \in \mathcal{F}_S$, for which $g(S) = k_S(S, s_0) - k_S(S, s_1)$ and $\|g\|_{\mathcal{F}_S}^2 = \|k_S(s_0, \cdot) - k_S(s_1, \cdot)\|_{\mathcal{F}_S}^2$. $\square$

SM1.9. Proof of Corollary 3.10 (Spectral control of the conditional MMD integral).

Proof. Since k_S is bounded, T_S is a self-adjoint, positive, trace-class operator on $L^2(P_S)$; let $(\lambda_\ell, \psi_\ell)_{\ell \geq 1}$ be its eigensystem with $\lambda_\ell \geq 0$ and $\{\psi_\ell\}$ orthonormal in $L^2(P_S)$, and recall the Mercer expansion $k_S(s, t) = \sum_{\ell \geq 1} \lambda_\ell \psi_\ell(s) \psi_\ell(t)$ in $L^2(P_S \otimes P_S)$. By Theorem 3.8,

$$\text{HSIC}(Z, S) = \|\mathbb{E}_S[\Delta(S) \otimes k_S(S, \cdot)]\|_{\mathcal{F}_Z \otimes \mathcal{F}_S}^2.$$

Substituting the Mercer expansion and writing $\Delta_\ell := \mathbb{E}_S[\Delta(S) \psi_\ell(S)] = \int \Delta(s) \psi_\ell(s) dP_S(s) \in \mathcal{F}_Z$ gives, by orthonormality of $\{\psi_\ell\}$,

$$\mathbb{E}_S[\Delta(S) \otimes k_S(S, \cdot)] = \sum_{\ell \geq 1} \lambda_\ell \Delta_\ell \otimes \psi_\ell, \quad \text{HSIC}(Z, S) = \sum_{\ell \geq 1} \lambda_\ell \|\Delta_\ell\|_{\mathcal{F}_Z}^2,$$

which is the first identity. For any m with $\lambda_m > 0$, since $\lambda_\ell \geq \lambda_m$ for $\ell \leq m$ and every term is nonnegative,

$$\sum_{\ell=1}^m \|\Delta_\ell\|_{\mathcal{F}_Z}^2 \leq \frac{1}{\lambda_m} \sum_{\ell=1}^m \lambda_\ell \|\Delta_\ell\|_{\mathcal{F}_Z}^2 \leq \frac{1}{\lambda_m} \text{HSIC}(Z, S).$$

Because $\{\psi_\ell\}$ is orthonormal in $L^2(P_S)$, the $\mathcal{F}_Z$ -valued coefficients of Δ are Δ_ℓ , so $\|P_m \Delta\|_{L^2(P_S; \mathcal{F}_Z)}^2 = \sum_{\ell=1}^m \|\Delta_\ell\|_{\mathcal{F}_Z}^2$, giving the projected bound. Finally, P_m is an orthogonal projection on $L^2(P_S; \mathcal{F}_Z)$, so $\|\Delta\|^2 = \|P_m \Delta\|^2 + \|(I - P_m) \Delta\|^2 \leq \lambda_m^{-1} \text{HSIC}(Z, S) + \rho_m^2$; the identity $\|\Delta\|_{L^2(P_S; \mathcal{F}_Z)}^2 = \mathbb{E}_S[\|\Delta(S)\|_{\mathcal{F}_Z}^2] = \mathbb{E}_S[\text{MMD}_{k_Z}^2(P_{Z|S}, P_Z)]$ uses $\|\Delta(s)\|_{\mathcal{F}_Z} = \text{MMD}_{k_Z}(P_{Z|S=s}, P_Z)$. $\square$

SM1.10. Proof of Corollary 3.11 (Spectral control of GDP for RKHS heads).

Proof. By the reproducing property, $m_f(s) = \mathbb{E}[f(Z) \mid S = s] - \mathbb{E}[f(Z)] = \langle f, \Delta(s) \rangle_{\mathcal{F}_Z}$, so $m_f \in L^2(P_S)$ with $L^2(P_S)$ coefficients $\int m_f(s) \psi_\ell(s) dP_S(s) = \langle f, \Delta_\ell \rangle_{\mathcal{F}_Z}$, where $\Delta_\ell := \mathbb{E}_S[\Delta(S) \psi_\ell(S)]$ as in the proof of Corollary 3.10. Hence, by Cauchy–Schwarz and that corollary,

$$\|P_m m_f\|_{L^2(P_S)}^2 = \sum_{\ell=1}^m \langle f, \Delta_\ell \rangle_{\mathcal{F}_Z}^2 \leq \|f\|_{\mathcal{F}_Z}^2 \sum_{\ell=1}^m \|\Delta_\ell\|_{\mathcal{F}_Z}^2 \leq \frac{\|f\|_{\mathcal{F}_Z}^2}{\lambda_m} \text{HSIC}(Z, S).$$

Since P_m is an orthogonal projection on $L^2(P_S)$, $\|m_f\|_{L^2(P_S)}^2 = \|P_m m_f\|^2 + \|(I - P_m)m_f\|^2 \leq \lambda_m^{-1} \|f\|_{\mathcal{F}_Z}^2 \text{HSIC}(Z, S) + r_{m,f}^2$. Finally, by Jensen’s inequality $\Delta_{\text{GDP}}(f) = \mathbb{E}_S|m_f(S)| \leq (\mathbb{E}_S m_f(S)^2)^{1/2} = \|m_f\|_{L^2(P_S)}$, so $\Delta_{\text{GDP}}(f)^2 \leq \lambda_m^{-1} \|f\|_{\mathcal{F}_Z}^2 \text{HSIC}(Z, S) + r_{m,f}^2$. $\square$

SM1.11. Proof of Theorem 3.12 (Empirical conditional-gap control by HSIC, projected version) and Corollary 3.13.

Proof of Theorem 3.12. We work with empirical quantities throughout. Denote by $\hat{d}_{S_i} := k_Z(Z_i, \cdot) - \hat{\mu}_Z$ the empirical centered embedding at $S = S_i$, with $\hat{\mu}_Z = n^{-1} \sum_{j=1}^n k_Z(Z_j, \cdot)$, and write $\delta_f := (\delta_{f,1}, \dots, \delta_{f,n})^\top$ for the centered prediction-deviation vector.

Step 1: Spectral identity for $\widehat{\text{HSIC}}$ using positive eigenpairs only. Let $K \in \mathbb{R}^{n \times n}$ have entries $K_{ij} = k_Z(Z_i, Z_j)$ and $\tilde{K} = HKH$. The biased V-statistic estimator is

$$\widehat{\text{HSIC}}(Z, S) = \frac{1}{n^2} \text{tr}(KHLH) = \frac{1}{n^2} \text{tr}(\tilde{K} \tilde{L}).$$

Since L is symmetric positive semi-definite and $H = H^\top$ with $H^2 = H$, $\tilde{L} = HLH$ is symmetric PSD. Let $\hat{\lambda}_1 \geq \dots \geq \hat{\lambda}_r > 0$ be the positive eigenvalues of $n^{-1} \tilde{L}$ with associated orthonormal eigenvectors $\hat{e}_1, \dots, \hat{e}_r \in \mathbb{R}^n$, where $r = \text{rank}(\tilde{L})$. Then

$$\frac{1}{n} \tilde{L} = \sum_{j=1}^r \hat{\lambda}_j \hat{e}_j \hat{e}_j^\top, \quad P_{\tilde{L}} = \sum_{j=1}^r \hat{e}_j \hat{e}_j^\top.$$

Substituting and using $\widehat{\text{HSIC}} = n^{-1} \text{tr}(\tilde{K} \cdot n^{-1} \tilde{L})$,

$$(SM1.4) \quad \widehat{\text{HSIC}}(Z, S) = \frac{1}{n} \sum_{j=1}^r \hat{\lambda}_j \cdot \hat{e}_j^\top \tilde{K} \hat{e}_j.$$

Step 2: Lower bound via the smallest positive eigenvalue. Since $\tilde{K} = HKH$ is PSD and $\hat{\lambda}_S = \min_{1 \leq j \leq r} \hat{\lambda}_j > 0$, every term in (SM1.4) satisfies $\hat{\lambda}_j \cdot \hat{e}_j^\top \tilde{K} \hat{e}_j \geq \hat{\lambda}_S \cdot \hat{e}_j^\top \tilde{K} \hat{e}_j$. Therefore

$$(SM1.5) \quad \widehat{\text{HSIC}}(Z, S) \geq \frac{\hat{\lambda}_S}{n} \sum_{j=1}^r \hat{e}_j^\top \tilde{K} \hat{e}_j = \frac{\hat{\lambda}_S}{n} \text{tr}(P_{\tilde{L}} \tilde{K} P_{\tilde{L}}),$$

where we used $\sum_{j=1}^r \hat{e}_j^\top \tilde{K} \hat{e}_j = \text{tr}(P_{\tilde{L}} \tilde{K} P_{\tilde{L}})$ (since $P_{\tilde{L}} = \sum_{j=1}^r \hat{e}_j \hat{e}_j^\top$, $P_{\tilde{L}}^2 = P_{\tilde{L}}$, and the cyclic-trace identity).

Step 3: Bound the projected prediction deviation by $\text{tr}(P_{\tilde{L}}\tilde{K}P_{\tilde{L}})$. By the reproducing property, $\hat{\delta}_{f,i} = \langle f, \hat{d}_{S_i} \rangle_{\mathcal{F}_{\mathcal{Z}}}$, so the vector $\boldsymbol{\delta}_f$ has entries $(\boldsymbol{\delta}_f)_i = \langle f, \hat{d}_{S_i} \rangle_{\mathcal{F}_{\mathcal{Z}}}$. For any unit vector $u \in \mathbb{R}^n$, the linear combination $\sum_i u_i \hat{d}_{S_i} \in \mathcal{F}_{\mathcal{Z}}$ and $u^\top \boldsymbol{\delta}_f = \langle f, \sum_i u_i \hat{d}_{S_i} \rangle_{\mathcal{F}_{\mathcal{Z}}}$. By Cauchy-Schwarz in $\mathcal{F}_{\mathcal{Z}}$,

$$(u^\top \boldsymbol{\delta}_f)^2 \leq \|f\|_{\mathcal{F}_{\mathcal{Z}}}^2 \left\| \sum_i u_i \hat{d}_{S_i} \right\|_{\mathcal{F}_{\mathcal{Z}}}^2 = \|f\|_{\mathcal{F}_{\mathcal{Z}}}^2 u^\top \tilde{K} u,$$

where the last equality uses $\langle \hat{d}_{S_i}, \hat{d}_{S_j} \rangle_{\mathcal{F}_{\mathcal{Z}}} = \tilde{K}_{ij}$. Applying this with $u = \hat{e}_j$ for $j = 1, \dots, r$ and summing,

$$\sum_{j=1}^r (\hat{e}_j^\top \boldsymbol{\delta}_f)^2 \leq \|f\|_{\mathcal{F}_{\mathcal{Z}}}^2 \sum_{j=1}^r \hat{e}_j^\top \tilde{K} \hat{e}_j = \|f\|_{\mathcal{F}_{\mathcal{Z}}}^2 \text{tr}(P_{\tilde{L}} \tilde{K} P_{\tilde{L}}).$$

The left-hand side equals $\|P_{\tilde{L}} \boldsymbol{\delta}_f\|_2^2 = \sum_{i=1}^n (P_{\tilde{L}} \boldsymbol{\delta}_f)_i^2$ since $\{\hat{e}_j\}_{j=1}^r$ is an orthonormal basis for the column span of $\tilde{L}$. Dividing by n ,

$$(SM1.6) \quad \frac{1}{n} \sum_{i=1}^n (P_{\tilde{L}} \boldsymbol{\delta}_f)_i^2 \leq \frac{\|f\|_{\mathcal{F}_{\mathcal{Z}}}^2}{n} \text{tr}(P_{\tilde{L}} \tilde{K} P_{\tilde{L}}).$$

Step 4: Combine. Combining (SM1.5) (which gives $\text{tr}(P_{\tilde{L}} \tilde{K} P_{\tilde{L}}) \leq n \widehat{\text{HSIC}}/\hat{\lambda}_S$) with (SM1.6),

$$\frac{1}{n} \sum_{i=1}^n (P_{\tilde{L}} \boldsymbol{\delta}_f)_i^2 \leq \frac{\|f\|_{\mathcal{F}_{\mathcal{Z}}}^2}{\hat{\lambda}_S} \widehat{\text{HSIC}}(Z, S),$$

which is (3.5). $\square$

Proof of Corollary 3.13. If $\text{rank}(\tilde{L}) = n - 1$, then the null space of $\tilde{L}$ is one-dimensional and contains $\mathbf{1}$ (since $\tilde{L}\mathbf{1} = HLH\mathbf{1} = 0$), so it equals $\text{span}\{\mathbf{1}\}$ and $P_{\tilde{L}}$ is the orthogonal projection onto its complement. The centered vector $\boldsymbol{\delta}_f$ satisfies $\mathbf{1}^\top \boldsymbol{\delta}_f = \sum_i (f(Z_i) - \bar{f}_n) = 0$, hence lies in the column span of $\tilde{L}$, so $P_{\tilde{L}} \boldsymbol{\delta}_f = \boldsymbol{\delta}_f$ and $\sum_i (P_{\tilde{L}} \boldsymbol{\delta}_f)_i^2 = \sum_i \hat{\delta}_{f,i}^2$. The first inequality of the corollary then follows from (3.5). The second (absolute-gap) inequality is Jensen applied to $x \mapsto x^2$:

$$\hat{\Delta}_f^2 = \left(\frac{1}{n} \sum_i |\hat{\delta}_{f,i}| \right)^2 \leq \frac{1}{n} \sum_i \hat{\delta}_{f,i}^2 \leq \frac{\|f\|_{\mathcal{F}_{\mathcal{Z}}}^2}{\hat{\lambda}_S} \widehat{\text{HSIC}}(Z, S). \quad \square$$

The population counterpart of this finite-sample spectral control is Corollary 3.10 of the main paper, which handles the compact kernel operator on $\mathcal{S}$ through its spectral tail ρ_m without assuming a spectral gap.

SM2. Additional Notation and Technical Lemmas.

LEMMA SM2.1 (Fubini's theorem for RKHS-valued integrals). *Let k be a bounded kernel, $\sup_{x \in \mathcal{X}} k(x, x) \leq \kappa < \infty$, with RKHS $\mathcal{F}$. For any random variable X on $\mathcal{X}$, the mean embedding $\mu_X = \mathbb{E}[k(X, \cdot)]$ exists as a Bochner integral in $\mathcal{F}$. Moreover, for any $\mathcal{F}$ -valued Bochner-integrable random element U and any $v \in \mathcal{F}$, $\langle \mathbb{E}[U], v \rangle_{\mathcal{F}} = \mathbb{E}[\langle U, v \rangle_{\mathcal{F}}]$; in particular $\langle \mu_X, v \rangle_{\mathcal{F}} = \mathbb{E}[v(X)]$.*

Proof. By the reproducing property,

$$\|k(x, \cdot)\|_{\mathcal{F}} = \sqrt{k(x, x)} \leq \sqrt{\kappa}, \quad \text{so} \quad \mathbb{E}[\|k(X, \cdot)\|_{\mathcal{F}}] \leq \sqrt{\kappa} < \infty,$$

guaranteeing existence of the Bochner integral. The exchange of expectation and inner product follows from continuity of the inner product together with Fubini's theorem. $\square$

SM2.1. Discrete sensitive attribute: coincidence of the two formulations. When S takes finitely many values, the joint and conditional-integral formulations of representation-level fairness reduce to the same aggregated two-sample MMD object, which motivates studying the continuous case in the main text.

LEMMA SM2.2 (Discrete sensitive attribute coincidence). *Use the notation of the main paper: $\mathcal{F}_Z$ is the RKHS on Z with characteristic kernel k_Z , $\mu_Z := \mathbb{E}[k_Z(Z, \cdot)] \in \mathcal{F}_Z$ is the marginal mean embedding, and $\mu_{Z|s_k} := \mathbb{E}[k_Z(Z, \cdot) \mid S = s_k] \in \mathcal{F}_Z$ is the conditional mean embedding. Let $\mathcal{S} = \{s_1, \dots, s_K\}$ be finite with $\pi_k := \Pr(S = s_k) > 0$, let k_S be the strictly positive-definite sensitive-attribute kernel of the main paper, write $\delta_k := \mu_{Z|s_k} - \mu_Z$, and let $\mathcal{I}_{\text{MMD}^2}(Z; S) := \mathbb{E}_S[\text{MMD}^2(P_{Z|S}, P_Z)]$ be the conditional-integral functional with local discrepancy $d = \text{MMD}^2$. Then*

$$\text{HSIC}(Z, S) = \sum_{k, k'=1}^K \pi_k \pi_{k'} k_S(s_k, s_{k'}) \langle \delta_k, \delta_{k'} \rangle_{\mathcal{F}_Z}.$$

$\text{HSIC}(Z, S)$ and $\mathcal{I}_{\text{MMD}^2}(Z; S)$ both vanish if and only if $P_{Z|S=s_k} = P_Z$ for every k ; both reduce to aggregated two-sample MMDs.

Proof. Expand HSIC via $\mu_{P_{Z,S}} - \mu_{P_Z \otimes P_S} = \mathbb{E}_S[(\mu_{Z|S} - \mu_Z) \otimes k_S(S, \cdot)]$ and integrate the finite measure on $\mathcal{S}$. $\square$

SM2.2. Bandwidth scaling of the spectral constant.

Remark SM2.3 (Bandwidth scaling of the spectral constant). With the normalization $\hat{\lambda}_S = \lambda_{\min}^+(n^{-1}\tilde{L})$, the spectral constant reflects the conditioning of the centered Gram matrix on S . As the sensitive-kernel bandwidth tends to zero and $L \rightarrow I_n$, $\tilde{L} \rightarrow H$ (whose positive eigenvalues are all 1), and $\hat{\lambda}_S \rightarrow 1/n$. As the bandwidth tends to infinity and $L \rightarrow \mathbf{1}\mathbf{1}^\top$, $\tilde{L} \rightarrow 0$ and $\hat{\lambda}_S \rightarrow 0$. Very large bandwidths therefore make the conditional-gap bound loose, while extremely small bandwidths give a finite-sample point-mass notion of conditioning; the median heuristic selects an intermediate regime. Standard results on Gaussian-kernel Gram matrices for samples from densities with compact support (e.g., [Koltchinskii and Giné, 2000](#)) confirm a polynomial dependence on σ between these extremes; in our experiments at $n = 1500$ this yields $\hat{\lambda}_S \sim 10^{-9}$ – 10^{-7} across the bandwidth sweep (Section [SM8.3](#)).

SM3. Regularized empirical kHGR. This appendix records a regularized empirical bound linking $\widehat{\text{HSIC}}$ to a Tikhonov-regularized empirical kHGR. The unregularized empirical kHGR requires inverting empirical covariance operators with arbitrarily small positive eigenvalues, so a meaningful finite-sample statement is naturally regularized; see [Fukumizu et al. \(2007\)](#) for the kCCA interpretation. The FRHSIC guarantees in the main text do not depend on this result.

Empirical covariance operators. Define the empirical centered cross-covariance operator $\hat{\Sigma}_{ZS} : \mathcal{F}_S \rightarrow \mathcal{F}_Z$ by

$$\hat{\Sigma}_{ZS} = \frac{1}{n} \sum_{i=1}^n (k_Z(Z_i, \cdot) - \hat{\mu}_Z) \otimes (k_S(S_i, \cdot) - \hat{\mu}_S),$$

where $\hat{\mu}_Z, \hat{\mu}_S$ are the empirical mean embeddings, and define $\hat{\Sigma}_{ZZ}, \hat{\Sigma}_{SS}$ analogously. All three operators are finite-rank (rank at most n), hence Hilbert–Schmidt without further assumptions on the kernels. By a direct expansion, $\widehat{\text{HSIC}}(Z, S) = \|\hat{\Sigma}_{ZS}\|_{\text{HS}}^2$.

Regularized empirical *kHGR*. For ridge parameters $\eta_Z, \eta_S > 0$, define the Tikhonov-regularized empirical *kHGR*

$$\widehat{\text{kHGR}}_{\eta_Z, \eta_S}(Z, S) := \|(\hat{\Sigma}_{ZZ} + \eta_Z I)^{-1/2} \hat{\Sigma}_{ZS} (\hat{\Sigma}_{SS} + \eta_S I)^{-1/2}\|_{\text{op}}.$$

PROPOSITION SM3.1 (Regularized empirical sandwich bound). Let k_Z, k_S be characteristic kernels and let $\eta_Z, \eta_S > 0$. Then

$$(\text{SM3.1}) \quad \widehat{\text{kHGR}}_{\eta_Z, \eta_S}(Z, S)^2 \leq \frac{\widehat{\text{HSIC}}(Z, S)}{\eta_Z \eta_S}.$$

Proof. Set $A = (\hat{\Sigma}_{ZZ} + \eta_Z I)^{-1/2}$, $B = \hat{\Sigma}_{ZS}$, $C = (\hat{\Sigma}_{SS} + \eta_S I)^{-1/2}$. Each operator $\hat{\Sigma}_{ZZ}, \hat{\Sigma}_{SS}$ is positive semi-definite, so $\hat{\Sigma}_{ZZ} + \eta_Z I \succeq \eta_Z I$ and $\hat{\Sigma}_{SS} + \eta_S I \succeq \eta_S I$. Consequently

$$\|A\|_{\text{op}} \leq 1/\sqrt{\eta_Z}, \quad \|C\|_{\text{op}} \leq 1/\sqrt{\eta_S}.$$

By the operator-norm $\leq$ Hilbert–Schmidt-norm inequality and the standard submultiplicative bound $\|ABC\|_{\text{op}} \leq \|A\|_{\text{op}} \|B\|_{\text{HS}} \|C\|_{\text{op}}$ (valid because the Hilbert–Schmidt norm dominates the operator norm),

$$\widehat{\text{kHGR}}_{\eta_Z, \eta_S}(Z, S) = \|ABC\|_{\text{op}} \leq \|A\|_{\text{op}} \|B\|_{\text{HS}} \|C\|_{\text{op}} \leq \frac{\|\hat{\Sigma}_{ZS}\|_{\text{HS}}}{\sqrt{\eta_Z \eta_S}}.$$

Squaring and using $\widehat{\text{HSIC}}(Z, S) = \|\hat{\Sigma}_{ZS}\|_{\text{HS}}^2$ gives (SM3.1). $\square$

The proposition makes no claim about the unregularized limit $\eta_Z, \eta_S \downarrow 0$, for which the spectrum of the empirical covariance operators must be controlled separately. Population-level zero-equivalence of HSIC and *kHGR* under characteristic kernels is independent of this regularization choice.

SM4. Experimental Details.

SM4.1. Datasets.

Adult Income. The UCI Adult dataset contains $n \approx 30,000$ records (after removing missing values). Features include numeric variables and one-hot encoded categorical ($d = 107$ after encoding). The target is whether income exceeds \$50K/year. The sensitive attribute is age (continuous, range 17–90). All features and the sensitive attribute are min-max scaled to $[0, 1]$.

ACS Income. The ACS Income dataset (Ding et al., 2021) is drawn from the 2018 American Community Survey (California) via the folktabs package. We subsample $n = 20,000$ individuals. The target is whether income exceeds \$50K/year. The sensitive attribute is age (continuous, range 17–94). Features include 9 demographic and employment variables.

MEPS. The Medical Expenditure Panel Survey (Panel 19, 2015) (Romano et al., 2020) contains $n \approx 13,000$ respondents after removing records with missing values (negative sentinel codes). The target is healthcare utilization ≥ 10 visits (binary classification). The sensitive attribute is age (continuous, range 17–85). Features include 19 variables covering demographics, health conditions, insurance status, and income.

Communities & Crime. The UCI Communities & Crime dataset contains $n \approx 2,000$ communities. The target is violent crimes per population (regression). The sensitive attribute is racial composition (continuous). Columns with $>20\%$ missing values are dropped; remaining missing values are imputed with column medians.

COMPAS. The ProPublica COMPAS dataset contains $n \approx 6,000$ defendants after standard filtering (Angwin et al., 2016). The target is two-year recidivism (binary classification). The sensitive attribute is age (continuous). Features include prior counts, charge degree (one-hot), sex, and race (one-hot), yielding $d = 16$ features.

SM4.2. Model Architecture. Following Kong et al. (2025), the encoder h is a 2-layer MLP: $d_X \rightarrow 50 \rightarrow 50$ with SELU activations (hidden dimension $H = 50$, output representation dimension $Z = 50$). The prediction head f is a linear layer: $50 \rightarrow 1$ (no hidden layers). This architecture matches FREM exactly, ensuring a fair comparison. The adversary in LAFTR and ADV uses the same 2-layer structure ($50 \rightarrow 50 \rightarrow n_{\text{out}}$) with SELU activation.

SM4.3. Training Details. All models are trained for 200 epochs with the Adam optimizer (learning rate 10^{-3} , default $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$, weight decay 0), batch size 256, and the standard PyTorch default initialization. For FRHSIC, kernel bandwidths are set via the median heuristic: σ_S is computed once on the training data, and σ_Z is recomputed every 20 epochs on the current mini-batch of encoded representations. For baselines (FREM, Reg-GDP, etc.), we use fixed $\sigma_Z = 1.0$ for the Z -kernel after min-max scaling to $[0, 1]$, following Kong et al. (2025); the additional S -smoothing bandwidths used by FREM ($\gamma = 0.5$) and Reg-GDP (Nadaraya–Watson bandwidth 0.2) are taken from their source publications and described per-method below. We do not modify per-method bandwidth conventions, ensuring each method is evaluated as published.

Train/validation/test splits. For all real datasets we use an 80%/20% random train/test split. Hyperparameter selection (i.e., choosing λ along the Pareto frontier) is performed using the same training split as the model fit; for each method and each dataset we sweep λ over the full grid and report results across the resulting Pareto frontier (no separate validation split is used to pick a single λ , since the goal of the comparison is to characterize the fairness–accuracy tradeoff curve, not to choose one operating point). We repeat 5 times with different random splits and report mean $\pm$ standard deviation on the held-out test set.

Regularization-strength (λ) grids. For all methods on all real datasets, we sweep

$$\lambda \in \{0.1, 1.0, 10.0, 100.0, 500.0\}.$$

For the Equal Opportunity and multi-sensitive-attribute experiments (Section SM5) the grid is $\lambda \in \{0.1, 1.0, 10.0, 50.0\}$.

SM4.4. Baseline Implementations and Hyperparameter Grids.

Reg-GDP. We implement the GDP regularizer from Jiang et al. (2022) using kernel-smoothed conditional expectations on a grid of 30 evenly-spaced points over the range of S . Reg-GDP uses two distinct bandwidths: a Gaussian Z -kernel bandwidth $\sigma_Z = 1.0$ (matching the FREM convention of Kong et al. (2025) on min-max-scaled features) and a Nadaraya–Watson smoothing bandwidth 0.2 on S for the conditional expectation; both follow the values reported by Jiang et al. (2022).

FREM. We implement the weighted EIPM estimator from Kong et al. (2025) using the MMD discriminator. Kernel-smoothed weights are computed with a Gaussian kernel on S with bandwidth $\gamma = 0.5$ (following Kong et al. (2025)). For computational tractability we subsample $n_{\text{anchor}} = 32$ anchor points per batch for the EIPM computation; see Section SM4.5 below for details.

LAFTR. We bin the continuous sensitive attribute into 4 quartiles and apply the adversarial FRL method of Madras et al. (2018). The adversary is trained with

alternating gradient steps (one adversary step per encoder step). The adversary is a 2-layer MLP ($50 \rightarrow 50 \rightarrow 4$) with SELU activations.

ADV. The continuous-target adversary of Grari et al. (2022), with the same 2-layer architecture as LAFTR but with a single scalar output (predicting S); trained with one adversary step per encoder step.

MMD-binned. Bins continuous S into $n_{\text{bins}} = 10$ equal-width bins and applies pairwise MMD across bins; bin centers serve as the smoothing grid.

dCor. We use the unbiased empirical distance correlation of Székely et al. (2007) as the fairness penalty; no further hyperparameters beyond λ .

SM4.5. FREM 32-anchor subsampling. The FREM EIPM objective of Kong et al. (2025) computes, at each mini-batch, an integral over s of a weighted MMD term:

$$\widehat{\text{EIPM}}_n(Z; S) = \int_S \widehat{\text{MMD}}^2(\hat{P}_{Z|S=s}, \hat{P}_Z) d\hat{P}_S(s).$$

Estimating this expectation by Monte Carlo with all n training points as anchors would require $O(n^3)$ work per batch (an $n \times n$ kernel matrix evaluated at each of n anchors). For computational tractability we instead draw $n_{\text{anchor}} = 32$ uniformly-random anchor points per mini-batch and average the MMD terms over those anchors. With batch size 256 this is a $32/256 = 12.5\%$ Monte Carlo subsample and reduces the per-batch cost to $O(n_{\text{anchor}} \cdot n^2)$. The number 32 was chosen to match the FREM implementation in our codebase (see `experiments/real_data.py`, function `train_frem`); we report results with $n_{\text{anchor}} = 32$ throughout.

FREM anchor-subsampling sensitivity (limitation). The 32-anchor subsampling we use for FREM is a computational approximation; an anchor-count sweep is the natural next step to characterize the sensitivity of FREM’s reported Pareto frontier to the anchor budget, and we plan to include such a sweep in a follow-up version of this work. The per-epoch wall-clock cost grows linearly with the anchor count, so the principal cost of the sweep is compute rather than implementation.

Baseline bandwidth sensitivity (limitation). The bandwidths reported in Section SM4 (FRHSIC median heuristic, FREM $\gamma = 0.5$ and $\sigma_Z = 1.0$, Reg-GDP Nadaraya–Watson bandwidth 0.2 and $\sigma_Z = 1.0$) are the values reported in the source publications. We retain published bandwidths in this paper to avoid favoring any particular method by retuning, but a full bandwidth-sensitivity sweep for the FREM and Reg-GDP baselines is the natural next step, and we plan to include such a sweep in a follow-up version of this work.

Paired significance test (limitation). We report mean and standard deviation across $n = 5$ random splits in Table 1 of the main paper but do not run a formal paired test (e.g., Wilcoxon signed-rank), since 5 paired observations is too few for reliable inference at conventional significance levels. Increasing the number of repeats to enable a credible paired comparison is the natural next step and we plan to include such an analysis in a follow-up version of this work.

SM4.6. Reproducibility checklist. We summarize the elements needed to reproduce the experiments end to end.

- **Random seed.** A single base seed `SEED = 42` is set at the top of each experiment script for `numpy`, `torch`, and the train/test split RNG. The 5 repeats use seeds `SEED, SEED + 1, ..., SEED + 4`. See `experiments/real_data.py:27` and `experiments/utils.py`.

- **Train/validation/test split.** 80%/20% random train/test split; no separate validation split (rationale above). Repeated 5 times with different splits.
- **Preprocessing.** Min-max scaling for the features and the sensitive attribute is fit on the *training split only* and then applied to the held-out test split (`experiments/real_data.py`, `run_single_dataset`). All cells of Table 1 in the main paper are reported under this leakage-free scaling protocol; the resulting numbers agree with our earlier leakage-affected numbers to within standard deviations on essentially every cell, with the most visible shifts in the MMD and ADV rows of Crime (where the small sample size $n \approx 2000$ amplifies any scaling difference). Categoricals are one-hot encoded; rows with missing values are dropped except in Communities & Crime, where columns with >20% missing are dropped and remaining missing values are imputed with column medians.
- **Model architecture.** Encoder h : 2-layer MLP $d_X \rightarrow 50 \rightarrow 50$ with SELU activations. Predictor f : linear $50 \rightarrow 1$. Adversaries (LAFTR / ADV / MMD-binned): 2-layer MLP $50 \rightarrow 50 \rightarrow n_{\text{out}}$ with SELU.
- **Optimizer.** Adam, learning rate 10^{-3} , $\beta_1 = 0.9, \beta_2 = 0.999$, $\varepsilon = 10^{-8}$, weight decay 0. PyTorch defaults.
- **Batch size and epochs.** Batch size 256, 200 epochs.
- **Kernel bandwidths.** FRHSIC: median heuristic for σ_S (fixed at start of training, on training data) and for σ_Z (recomputed every 20 epochs on the current mini-batch). Baselines: fixed $\sigma_Z = 1.0$ on min-max-scaled Z , FREM $\gamma = 0.5$, Reg-GDP NW bandwidth 0.2.
- **λ grids.** Main experiments: $\lambda \in \{0.1, 1.0, 10.0, 100.0, 500.0\}$. EO and multi-sensitive: $\lambda \in \{0.1, 1.0, 10.0, 50.0\}$.
- **Baseline hyperparameter grids.** See Section SM4, “Baseline Implementations”; all baselines use the same encoder, predictor, optimizer, batch size, and number of epochs as FRHSIC, varying only the per-method fairness-penalty hyperparameters listed above. The adversary step ratio (LAFTR, ADV) is fixed at 1:1.
- **FREM anchor count.** $n_{\text{anchor}} = 32$ uniformly-sampled anchors per mini-batch; see Section SM4.5.
- **Hardware.** All experiments were run on a single workstation with one NVIDIA RTX-class GPU; the experiment code falls back to CPU if no CUDA device is available (see the device selector in `experiments/utils.py`). Runtime numbers in Section 5 of the main paper are wall-clock per-epoch on this single device, averaged over 5 epochs after a warm-up epoch. Training the full λ -sweep across all methods and datasets fits within ~ 24 GPU-hours.
- **Software.** Python 3.11, PyTorch 2.x, NumPy 1.26, scikit-learn 1.3, folktables (for ACS Income). See `experiments/README.md` for the exact versions used at submission time.
- **Code availability.** All scripts to reproduce the figures and tables are in the `experiments/` directory of the public code repository (<https://github.com/Yijin911/FRHSIC>); each main-paper figure/table is generated by a single named script (e.g. `run_single_method.py`, `regen_pareto_mi.py`, `runtime.py`, `convergence.py`).

SM4.7. Pareto summary: lowest GDP within 1% of the unfair-baseline performance. For a single-number summary of the fairness-accuracy tradeoff that complements the matched-performance comparison in Section 5 of the main paper,

Table SM1 reports, for each (method, dataset) pair, the operating point on the λ -sweep that attains the *lowest GDP* subject to the constraint that the prediction performance is within 1% of the unfair baseline ($\text{Acc} \geq 0.99 \cdot \text{Acc}_{\text{Unfair}}$ for classification; $\text{MSE} \leq 1.01 \cdot \text{MSE}_{\text{Unfair}}$ for regression). Entries are read off the Pareto sweep underlying Figure 3 of the main paper. “—” denotes that no λ in our grid satisfies the constraint for that method on that dataset. Values are means across 5 random splits.

TABLE SM1

Pareto summary: (Acc / MSE, GDP) at the lowest-GDP operating point with prediction performance within 1% of the unfair baseline. Adult/ACS/MEPS/COMPAS report Acc; Crime reports $\text{MSE} \times 10^2$. “—” indicates no λ in the sweep satisfies the constraint. Read alongside Figure 3 of the main paper, since a tight 1% band can exclude operating points that are visually close on the frontier.

Method	Adult	ACS Income	MEPS	Crime	COMPAS
Unfair (ref.)	(.849, .749)	(.790, .514)	(.804, .626)	(1.91, .029)	(.654, .123)
FRHSIC (Ours)	(.845, .232)	(.792, .399)	(.803, .517)	(1.91, .029)	(.655, .107)
FREM	(.845, .167)	(.787, .274)	(.800, .317)	(1.87, .023)	(.648, .101)
Reg-GDP	(.842, .082)	(.787, .292)	(.801, .370)	(1.91, .025)	(.650, .079)
ADV	(.841, .466)	(.785, .458)	(.803, .480)	(1.92, .029)	(.649, .095)
MMD (binned)	(.847, .126)	(.783, .157)	(.800, .407)	—	(.651, .082)
LAFTR (binned)	(.842, .385)	(.787, .474)	(.803, .530)	(1.89, .029)	(.655, .122)
dCor	(.846, .194)	(.783, .203)	(.800, .487)	(1.92, .023)	(.653, .091)

Caveats. The 1% accuracy band is intentionally tight, and on some datasets (notably COMPAS, where multiple methods cluster near the unfair-baseline accuracy) the band excludes operating points that are visually close on the Pareto frontier (Figure 3). Reg-GDP can drive GDP to near zero on Adult and ACS, but only at accuracy losses well outside the 1% band; those points are visible on Figure 3 but do not appear in the table above. The single-number summary should therefore be read alongside the full Pareto curves rather than as a standalone ranking.

SM5. Additional Experimental Results.

SM5.1. Synthetic validation of the conditional-gap bound. On a controlled synthetic classification setup, enforcing HSIC drives GDP monotonically to zero at stable accuracy, confirming the conditional-gap bound of Theorem 3.12. We generate $n = 5000$ samples with $S \sim \text{Uniform}(0, 1)$, $X = (S + \epsilon_1, \epsilon_2)$ where $\epsilon_1 \sim \mathcal{N}(0, 0.09)$, $\epsilon_2 \sim \mathcal{N}(0, 1)$, and $Y \sim \text{Bernoulli}(1/(1 + e^{-X_1}))$. The encoder h is a 2-layer MLP (hidden dimension 64, representation dimension 8) and the prediction head f is linear; a smaller architecture than the real-data experiments isolates the effect of the HSIC penalty in a controlled setting. We train with objective $\mathcal{L}_{\text{CE}}(f \circ h) + \lambda \cdot \text{HSIC}(Z, S)$ for $\lambda \in \{0, 0.1, 0.5, 1, 5, 10, 50\}$. As λ increases, GDP decreases monotonically toward zero (Figure SM1a), GDP and HSIC decrease together in a monotone relationship consistent with the upper bound in Theorem 3.12 (Figure SM1c), and accuracy remains stable (around 0.62), indicating that the sensitive attribute contributes primarily to unfair discrimination rather than useful predictive signal in this setting.

SM5.2. Equal Opportunity and multiple sensitive attributes. The HSIC penalty extends to Equal Opportunity (Madras et al., 2018), which requires fairness conditional on the positive outcome $Y = 1$, by restricting the estimator to the n_1

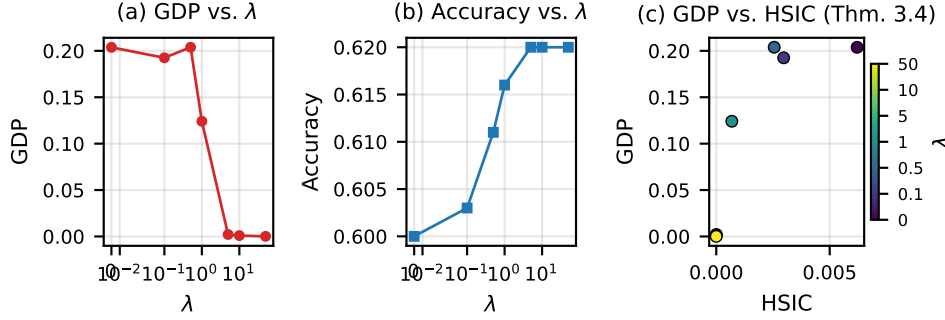


FIG. SM1. *Synthetic validation.* (a) GDP decreases as λ increases. (b) Accuracy remains stable across regularization strengths. (c) GDP vs. HSIC shows a monotone relationship, consistent with the conditional-gap-control interpretation of Theorem 3.12 (the plot displays monotonic dependence, not the LHS/RHS of the bound).

472 samples with $Y_i = 1$:

$$473 \quad \widehat{\text{HSIC}}_{\text{EO}}(Z, S) = \frac{1}{n_1^2} \text{tr}(K_1 H_1 L_1 H_1),$$

474 where K_1, L_1, H_1 are the kernel and centering matrices restricted to that subset. It
 475 extends to multiple continuous sensitive attributes by taking a product kernel on
 476 $\mathcal{S}_1 \times \mathcal{S}_2$ (Appendix SM5.3).

477 We evaluate the EO extension on Adult and COMPAS, comparing FRHSIC-DP
 478 (HSIC on all samples) to FRHSIC-EO (HSIC on $Y = 1$ subset only). Table SM2
 479 reports accuracy and EO-GDP (GDP restricted to the $Y = 1$ subset) across λ values.
 480 On COMPAS, the DP variant achieves comparable or better EO-GDP than the EO
 481 variant at similar accuracy across all λ values. On Adult, the EO variant achieves
 482 slightly lower EO-GDP at high λ (0.352 vs. 0.418 at $\lambda = 50$), but the DP variant
 483 simultaneously reduces full GDP from 0.889 to 0.240—a broader fairness guarantee.
 484 Overall, enforcing full demographic parity via HSIC provides a strong baseline for
 485 Equal Opportunity, with the dedicated EO variant offering marginal gains on larger
 486 datasets.

487 **SM5.3. Multiple Sensitive Attributes.** We test FRHSIC with two contin-
 488 uous sensitive attributes (age and hours-per-week) on Adult. Table SM3 compares
 489 two approaches: (1) a product kernel on joint (S_1, S_2) (FRHSIC-Joint), and (2) a
 490 sum of per-attribute HSIC terms (FRHSIC-Sum). Both approaches reduce GDP
 491 with respect to each attribute as λ increases while preserving accuracy, confirming
 492 that HSIC naturally extends to the multi-attribute setting without modification. The
 493 Joint approach achieves slightly lower GDP at $\lambda = 50$ (Age: 0.306 vs. 0.678; Hours:
 494 0.517 vs. 0.759), likely because the product kernel captures cross-attribute structure.
 495 Figure SM2 visualizes the Pareto frontiers.

496 SM6. Structural Causal Models.

497 **SM6.1. Structural Causal Models and Counterfactual Fairness.** Follow-
 498 ing Pearl (2009) and Kusner et al. (2017), a Structural Causal Model (SCM) over
 499 (X, S, Y) specifies structural equations

$$500 \quad S = U_S, \quad X = f_X(S, U_X), \quad Y = f_Y(X, S, U_Y),$$

TABLE SM2

Equal Opportunity results: FRHSIC-DP vs. FRHSIC-EO. EO-GDP is GDP restricted to $Y = 1$ samples. FRHSIC-DP achieves comparable EO fairness while also enforcing DP.

Dataset	λ	Method	Acc	EO-GDP ($\downarrow$)
Adult	0.1	DP	0.851	0.453
		EO	0.849	0.406
	1.0	DP	0.846	0.403
		EO	0.850	0.416
	10.0	DP	0.846	0.437
		EO	0.850	0.392
	50.0	DP	0.840	0.418
		EO	0.849	0.352
COMPAS	0.1	DP	0.659	0.067
		EO	0.662	0.070
	1.0	DP	0.662	0.070
		EO	0.662	0.075
	10.0	DP	0.657	0.064
		EO	0.655	0.065
	50.0	DP	0.599	0.033
		EO	0.659	0.076

TABLE SM3

Multiple sensitive attributes on Adult (age + hours-per-week). GDP is reported separately for each attribute. Both product-kernel (Joint) and sum-of-HSIC (Sum) approaches reduce GDP for both attributes as λ increases.

Method (λ)	Acc	GDP(Age) ($\downarrow$)	GDP(Hours) ($\downarrow$)
Unfair	0.843	0.837	1.142
FRHSIC-Joint ($\lambda = 1$)	0.844	0.779	1.029
FRHSIC-Joint ($\lambda = 10$)	0.846	0.553	0.857
FRHSIC-Joint ($\lambda = 50$)	0.851	0.306	0.517
FRHSIC-Sum ($\lambda = 1$)	0.847	0.753	1.055
FRHSIC-Sum ($\lambda = 10$)	0.848	0.485	0.826
FRHSIC-Sum ($\lambda = 50$)	0.849	0.678	0.759

with jointly independent exogenous noise U_X, U_S, U_Y . The counterfactual prediction $\hat{Y}_{S \leftarrow s'}$ is the value of $\hat{Y}$ obtained by replacing $S = U_S$ with $S = s'$ in the structural equations and propagating downstream. A predictor $\hat{Y}$ is *counterfactually fair* (Kusner et al., 2017) if, for every observable $(X = x, S = s)$ and every alternative s' ,

$$\hat{Y}_{S \leftarrow s'} = \hat{Y}_{S \leftarrow s} \quad \text{a.s. given } X = x, S = s.$$

SM7. Additional Theoretical Augmentations. This section collects two augmentations referenced from the main paper: a necessity-for-counterfactual-fairness result and a uniform-concentration bound that bridges training-sample HSIC to its population counterpart.

SM7.1. Necessity for Counterfactual Fairness. The joint-distribution route to independence also clarifies the relationship between HSIC-fairness and the

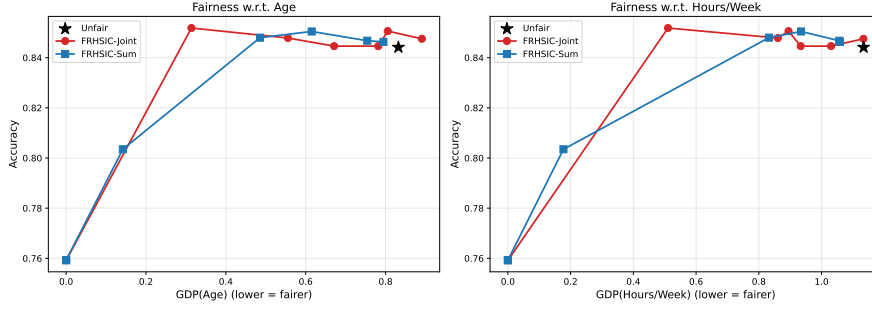


FIG. SM2. Multiple sensitive attributes: fairness–accuracy Pareto frontiers for FRHSIC-Joint (product kernel) and FRHSIC-Sum (sum of per-attribute HSIC), evaluated on Adult with age and hours-per-week as sensitive attributes.

causal-fairness literature.

PROPOSITION SM7.1 (HSIC-fairness as a necessary condition for counterfactual fairness). Assume the SCM in Section SM6.1 of this supplement. Suppose the encoder h is a deterministic function so that $Z = h(X) = h(f_X(S, U_X))$. If the prediction $\hat{Y} = f(Z)$ is counterfactually fair (Section SM6.1 of this supplement) for every f in a class $\mathcal{F}$ rich enough to identify equality of distributions in $\mathcal{Z}$ (e.g., $\mathcal{F}$ is the unit ball of an RKHS with characteristic kernel k_Z), then $\text{HSIC}(Z, S) = 0$.

Proof. Counterfactual fairness for every $f \in \mathcal{F}$ implies, in particular, that for all $s' \in \mathcal{S}$ in the support of S and all $f \in \mathcal{F}$,

$$\mathbb{E}[f(Z_{S \leftarrow s'})] = \mathbb{E}[f(Z)],$$

where $Z_{S \leftarrow s'} = h(f_X(s', U_X))$. Since $S = U_S$ and U_X are independent under the SCM, and the marginal of $Z_{S \leftarrow s'}$ over U_X has the same distribution as the marginal of Z given $S = s'$ (because intervening with $S = s'$ matches the conditional given $S = s'$ when S is exogenous (Pearl, 2009, Section 3.2.2)), we obtain

$$\mathbb{E}[f(Z) \mid S = s'] = \mathbb{E}[f(Z)] \quad \text{for } P_S\text{-a.e. } s'.$$

By Proposition 3.1 of the main paper, this is equivalent to $\text{HSIC}(Z, S) = 0$. $\square$

In words, if a representation can support counterfactually fair predictions for every (sufficiently rich) prediction head, the representation must already satisfy the observational-independence condition that HSIC enforces. The converse does not hold: $\text{HSIC}(Z, S) = 0$ ensures that the marginal distribution of $f(Z)$ does not depend on S , but counterfactual fairness is a strictly pointwise statement about individual-level interventions, which can fail when the structural dependence on S is observationally invisible (e.g., $Z = S \oplus U_X$ with binary S, U_X uniform yields $Z \perp S$ marginally but $Z_{S \leftarrow s'} \neq Z_{S \leftarrow s}$ pointwise). HSIC-fairness should therefore be understood as the strongest *observational* fairness consequence of counterfactual fairness, not as a substitute for it.

SM7.2. Uniform Concentration over the Encoder Class. The empirical Theorem 3.12 of the main paper controls the projected empirical second-moment conditional gap $\frac{1}{n} \sum_i (P_L \delta_f)_i^2$ through $\widehat{\text{HSIC}}(h(X), S)$ for a single encoder h (and, under the full-rank hypothesis of Corollary 3.13, the unprojected $\frac{1}{n} \sum_i \hat{\delta}_{f,i}^2$). In practice, h

is learned from the same data, so the bound must hold uniformly over a hypothesis class $\mathcal{H}$ of encoders. We invoke Corollary 23 of [Ni and Huo \(2024\)](#) (an HSIC specialization of their Theorem 12), which we adapt to the one-sided encoder class arising in FRHSIC. We do not reprove that result; this appendix only maps FRHSIC onto it.

Mapping FRHSIC to the prior uniform concentration result. First, $\text{HSIC}(Z, S)$ is the squared MMD between $P_{Z,S}$ and $P_Z \otimes P_S$ under the fixed product kernel $k((z, s), (z', s')) = k_Z(z, z') k_S(s, s')$, and $\widehat{\text{HSIC}}_n(h(X), S) = n^{-2} \text{tr}(K_h H L H)$ is exactly the empirical kernel-based two-sample statistic to which Theorem 12 of [Ni and Huo \(2024\)](#) applies, with $(Z', S') \sim P_Z \otimes P_S$. Second, the encoder class $\mathcal{H}$ acts only on the Z -coordinate, inducing the transformed paired samples $\{(h(X_i), S_i)\}$ and the encoder-indexed product-kernel class $\mathcal{K}_{\mathcal{H}}$. Third, the boundedness and Lipschitz hypotheses needed by [Ni and Huo \(2024\)](#) (Assumption 20) are the bounded-kernel and Lipschitz-feature-map conditions stated in Proposition [SM7.2](#), which k_Z , k_S , and $h \in \mathcal{H}$ satisfy under the standing assumptions. Fourth, bounded linear maps over a compact parameter set, and fixed-architecture MLPs with bounded weights, bounded inputs, and Lipschitz activations, have empirical Gaussian (equivalently, up to logarithmic factors, Rademacher) complexity of order $n^{-1/2}$, so the complexity term in the imported bound is controlled; growing-width or unregularized networks need not satisfy this. Data-dependent kernel selection, which would enlarge $\mathcal{K}_{\mathcal{H}}$ to a composite-kernel class, is outside the scope of this statement.

PROPOSITION SM7.2 (Uniform concentration of $\widehat{\text{HSIC}}$ over an encoder class). *Let $\mathcal{H}$ be a class of measurable encoders $h : \mathcal{X} \rightarrow \mathcal{Z}$. Suppose the kernels k_Z, k_S satisfy Assumption 20 of [Ni and Huo \(2024\)](#): they are bounded with $\sup_z k_Z(z, z) \leq \nu_Z$ and $\sup_s k_S(s, s) \leq \nu_S$, and the feature maps $z \mapsto k_Z(z, \cdot)$ and $s \mapsto k_S(s, \cdot)$ are Lipschitz with constants $\ell_Z, \ell_S > 0$, respectively (the constants $\nu_Z, \nu_S, \ell_Z, \ell_S$ are exactly those of Theorem [4.1](#) of the main paper). Let*

$$\hat{\mathcal{G}}_n(\mathcal{H}) := \mathbb{E}_{\xi} \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \langle \xi_i, h(X_i) \rangle \mid \{X_i\}_{i=1}^n \right]$$

denote the empirical Gaussian complexity of $\mathcal{H}$ ([Ni and Huo, 2024](#), Equation 3.5), where $\xi_1, \dots, \xi_n$ are i.i.d. $\mathcal{N}(0, I_{\dim(\mathcal{Z})})$ random vectors; write $\mathcal{G}_n(\mathcal{H}) := \mathbb{E}[\hat{\mathcal{G}}_n(\mathcal{H})]$ for its expectation, as in the main paper. Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\begin{aligned} \sup_{h \in \mathcal{H}} |\widehat{\text{HSIC}}(h(X), S) - \text{HSIC}(h(X), S)| &\leq 8\sqrt{2} \nu_Z \nu_S \sqrt{\frac{\log(2/\delta)}{n}} + \frac{4 \nu_Z \nu_S}{n} \\ &\quad + 48\sqrt{\pi} \max\{\nu_Z \ell_S, \nu_S \ell_Z\} \mathcal{G}_n(\mathcal{H}). \end{aligned}$$

Proof. Apply Corollary 23 of [Ni and Huo \(2024\)](#) to the two-sample statistic γ_k^2 on the product RKHS with reproducing kernel $k((z, s), (z', s')) = k_Z(z, z') k_S(s, s')$, identified with HSIC via $\text{HSIC}(Z, S) = \gamma_k^2((Z, S), (Z', S'))$ for $(Z', S') \sim P_Z \otimes P_S$ ([Ni and Huo, 2024](#), Equation 3). The bivariate function class $\mathcal{H} \times \{\text{id}_S\}$, where id_S is the identity on S , satisfies $\mathbb{E}[\mathcal{G}((\mathcal{H} \times \{\text{id}_S\})(\mathbf{X}, \mathbf{S}))] \leq \mathbb{E}[\hat{\mathcal{G}}_n(\mathcal{H})]$ by Lemma 19 of [Ni and Huo \(2024\)](#), since the second-coordinate class is a singleton: under the no-absolute-value convention adopted by [Ni and Huo \(2024, Section 3.1\)](#), $\mathcal{G}(\{\text{id}_S\}(\mathbf{S})) = \mathbb{E}_{\xi}[\frac{1}{n} \sum_i \langle \xi_i, S_i \rangle] = 0$ by symmetry of the Gaussian distribution. Substituting into the HSIC bound of Corollary 23 yields ([SM7.1](#)). $\square$

COROLLARY SM7.3 (Train-population HSIC bridge). *Under the assumptions of Proposition [SM7.2](#), for any encoder $\hat{h} \in \mathcal{H}$ obtained from n i.i.d. training samples and*

any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$(SM7.2) \quad \text{HSIC}(\hat{h}(X), S) \leq \widehat{\text{HSIC}}(\hat{h}(X), S) + \varepsilon_n(\delta),$$

where

$$\varepsilon_n(\delta) := 8\sqrt{2} \nu_Z \nu_S \sqrt{\frac{\log(2/\delta)}{n}} + \frac{4\nu_Z \nu_S}{n} + 48\sqrt{\pi} \max\{\nu_Z \ell_S, \nu_S \ell_Z\} \mathcal{G}_n(\mathcal{H})$$

is the uniform-concentration penalty from Proposition SM7.2.

Proof. By Proposition SM7.2, with probability at least $1 - \delta$,

$$\sup_{h \in \mathcal{H}} |\widehat{\text{HSIC}}(h(X), S) - \text{HSIC}(h(X), S)| \leq \varepsilon_n(\delta).$$

Specializing to the data-dependent $\hat{h} \in \mathcal{H}$, $\text{HSIC}(\hat{h}(X), S) - \widehat{\text{HSIC}}(\hat{h}(X), S) \leq \varepsilon_n(\delta)$, which rearranges to (SM7.2). $\square$

Remark SM7.4 (Train-test fairness bridge). Corollary SM7.3 delivers a train-test fairness guarantee when combined with Theorem 3.12 of the main paper applied on an independent test sample. Theorem 3.12 controls the projected empirical second moment $\frac{1}{n} \sum_i (P_{\tilde{L}} \delta_f)_i^2$, which under the full-rank hypothesis of Corollary 3.13 simplifies to $\frac{1}{n} \sum_i \hat{\delta}_{f,i}^2$ and, by Jensen’s inequality, dominates the squared empirical absolute gap for any prediction head $f \in \mathcal{F}_Z$. Concretely, given a test sample $\{(\tilde{Z}_i, \tilde{S}_i)\}_{i=1}^n$ independent of the training sample, Theorem 3.12 gives the deterministic bound

$$\frac{1}{n} \sum_{i=1}^n (P_{\tilde{L}^{\text{test}}} \delta_f^{\text{test}})_i^2 \leq \frac{\|f\|_{\mathcal{F}_Z}^2}{\tilde{\lambda}_S} \widehat{\text{HSIC}}^{\text{test}}(\hat{h}(X), S),$$

where $\widehat{\text{HSIC}}^{\text{test}}$, $\tilde{\lambda}_S$, and the projection $P_{\tilde{L}^{\text{test}}}$ are computed on the test sample. Applying Proposition SM7.2 once on training and once pointwise on test (the latter being the standard single-encoder $O(n^{-1/2})$ deviation, which is dominated by ε_n),

$$\widehat{\text{HSIC}}^{\text{test}}(\hat{h}(X), S) \leq \text{HSIC}(\hat{h}(X), S) + \varepsilon_n(\delta) \leq \widehat{\text{HSIC}}^{\text{train}}(\hat{h}(X), S) + 2\varepsilon_n(\delta)$$

with probability at least $1 - 2\delta$. Hence, with the same probability,

$$\frac{1}{n} \sum_{i=1}^n (P_{\tilde{L}^{\text{test}}} \delta_f^{\text{test}})_i^2 \leq \frac{\|f\|_{\mathcal{F}_Z}^2}{\tilde{\lambda}_S} [\widehat{\text{HSIC}}^{\text{train}}(\hat{h}(X), S) + 2\varepsilon_n(\delta)].$$

This is the generalization statement for the FRHSIC fairness surrogate: training-sample HSIC, plus the uniform-concentration penalty, controls the held-out projected empirical second-moment conditional gap up to the test-sample spectral factor $1/\tilde{\lambda}_S$.

SM8. Additional Experiments.

SM8.1. Representation-level mutual information. As a representation-level diagnostic complementing the transfer experiment, FRHSIC tends to attain lower $\text{MI}(Z, S)$ than Reg-GDP at comparable prediction performance, consistent with the learned representation carrying less information about S rather than only making a single trained head fair. Figure SM3 reports $\text{MI}(Z, S)$ versus prediction performance across the λ sweep. We treat $\text{MI}(Z, S)$ as a diagnostic, not a primary

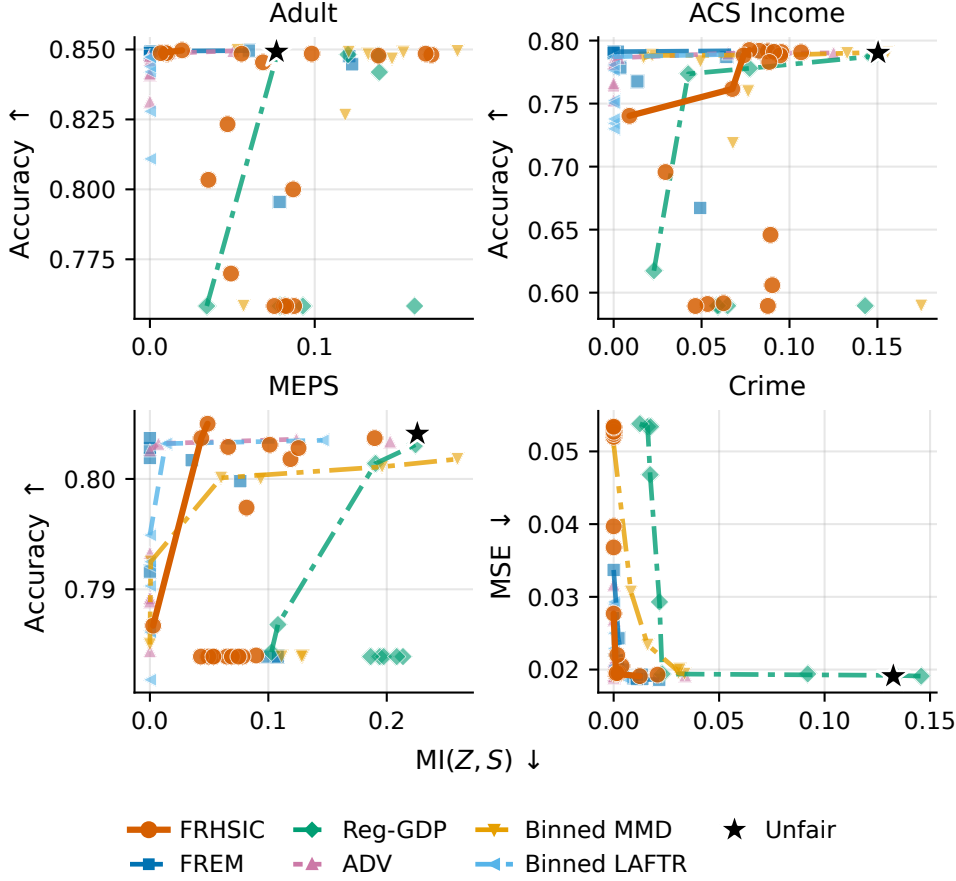


FIG. SM3. *FRHSIC* yields representations that leak less about S : it tends to attain lower $MI(Z, S)$ than *Reg-GDP* at comparable prediction performance, consistent with representation-level rather than single-head fairness. Axes are $MI(Z, S)$ (lower = fairer) versus performance across λ .

618 fairness audit: the KSG estimator can underestimate MI for high-dimensional learned
 619 representations whose dependence structure violates the smoothness assumptions
 620 underlying nearest-neighbor entropy estimation (Gretton et al., 2012), so we interpret
 621 it together with HSIC, GDP, and downstream performance. The adversarial methods
 622 illustrate this: ADV and LAFTR can report $MI \approx 0$ while their GDP remains high,
 623 which is inconsistent with $MI = 0 \Rightarrow Z \perp S \Rightarrow GDP = 0$ and indicates that adversar-
 624 ially trained representations, often concentrated on low-dimensional manifolds, are
 625 poorly suited to nearest-neighbor entropy estimation. COMPAS is omitted because
 626 the KSG estimator is unreliable on its low-dimensional, discrete-valued features.

627 **SM8.2. Transfer Experiment.** The key advantage of FRHSIC over Reg-GDP
 628 is *transferability*: since HSIC enforces $Z \perp S$, any downstream prediction head built
 629 on the learned representation is guaranteed to be fair. To test this empirically, we
 630 train representations using each method (at $\lambda = 10$), freeze the encoder, and fit four
 631 different downstream heads on the frozen representation: linear, 2-layer MLP, random
 632 forest, and SVM. We then measure GDP for each head.

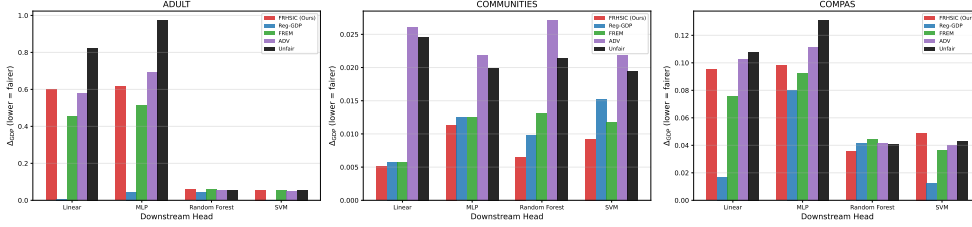


FIG. SM4. *Transfer experiment: GDP of different downstream prediction heads trained on frozen representations. In these experiments FRHSIC maintains comparably low GDP across heads, while Reg-GDP and FRM exhibit more head-dependent variation.*

Figure SM4 shows the results on all five datasets; we highlight two representative findings. On Crime ($\lambda = 100$), FRHSIC attains low GDP variance across heads (std = 0.001), comparable to or lower than Reg-GDP (0.002) and ADV (0.005). On Adult ($\lambda = 10$), FRHSIC reduces cross-head GDP variance from 0.348 (Unfair) to 0.244, a 30% reduction. Notably, Reg-GDP achieves lower variance (0.058) on Adult but at the cost of collapsing accuracy to 0.758 (Table 1 of the main paper). Among methods that preserve accuracy, FRHSIC provides the most consistent fairness across downstream heads. The MI results in Table 1 of the main paper provide complementary evidence: MI measures the total information about S in Z , bounding the discrimination achievable by *any* downstream head (including adversarial ones not tested here). FRHSIC’s low MI (0.008 on Adult, 0.015 on Crime) thus guarantees fairness even for worst-case downstream use, while Reg-GDP’s higher MI (0.017 on Adult, 0.056 on Crime) leaves room for adversarial exploitation despite its GDP = 0 for the training head.

SM8.3. Empirical Tightness of the Theorem 3.12 Bound. We empirically evaluate the tightness of the bound in Theorem 3.12 of the main paper on two real datasets (Adult and Crime; we omit COMPAS because age in this dataset takes only a small number of distinct integer values, so the centered Gram matrix $\tilde{L}$ on S is near-low-rank and $\hat{\lambda}_S$ is numerically unstable). For each dataset and each regularization strength λ , we compute (i) the empirical RHS $\|f\|^2 / \hat{\lambda}_S \cdot \widehat{\text{HSIC}}(Z, S)$ on the test split, where $\hat{\lambda}_S$ is the smallest positive eigenvalue of $n^{-1}\tilde{L}$, with $\tilde{L} = HLH$ the centered Gram matrix on S (eigenvalues below $10^{-6}\hat{\lambda}_{\max}$ are treated as numerical zero when identifying the smallest positive eigenvalue, mirroring standard practice for truncated/regularized kernel inverses); and (ii) the empirical LHS $\frac{1}{n} \sum_i \hat{\delta}_{f,i}^2$, averaged over RKHS test functions $f(\cdot) = k_Z(z_0, \cdot)$ for randomly sampled anchors z_0 . We plot both as a function of λ on log-log axes.

Figure SM5 shows the results. The bound is valid: the empirical RHS upper-bounds the LHS at every λ on both datasets. The vertical gap between the two curves can span several orders of magnitude and reflects the well-known looseness of HSIC-based bounds when $\hat{\lambda}_S$ is small (the multiplier $1/\hat{\lambda}_S$ is large because $\hat{\lambda}_S$ scales as a small fraction of the largest eigenvalue of $n^{-1}\tilde{L}$ for the Gaussian kernel at the median-heuristic bandwidth). Crucially, the empirical bound remains finite and non-vacuous — in sharp contrast with the population-level version, where the analogous constant collapses to zero for Gaussian kernels on continuous S and the bound becomes vacuous — so the empirical reformulation of Theorem 3.12 is quantitatively useful as a tractable surrogate.

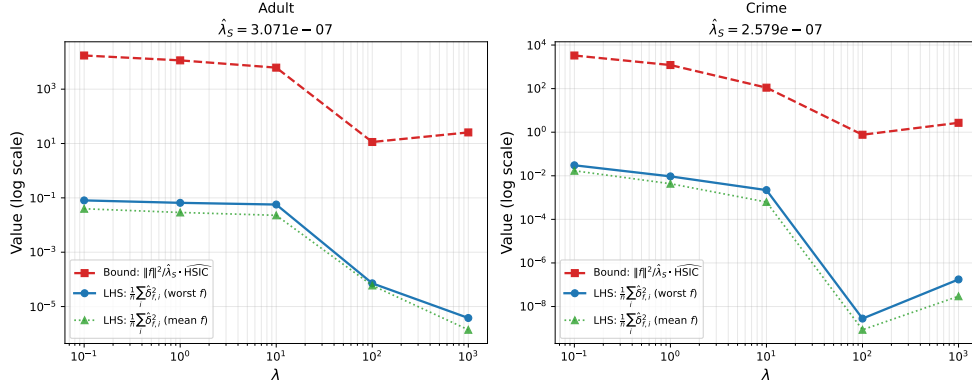


FIG. SM5. Empirical tightness of the bound from Theorem 3.12 of the main paper. Solid line: empirical LHS $\frac{1}{n} \sum_i \hat{\delta}_{f,i}^2$. Dashed line: empirical RHS $\|f\|^2 / \hat{\lambda}_S \cdot \widehat{\text{HSIC}}$. Validity: the RHS upper-bounds the LHS at every λ . The vertical gap can span several orders of magnitude and reflects the looseness of HSIC-based bounds when $\hat{\lambda}_S$ is small; the bound nonetheless remains finite and non-vacuous, in contrast to the population version, where the constant collapses to zero.

669 Bandwidth scaling of $\hat{\lambda}_S$. To validate the polynomial-scaling claim in
 670 Remark SM2.3 of the main paper, we additionally sweep the bandwidth
 671 $\sigma \in \{0.1, 0.3, 1.0, 3.0, 10.0\}$ and record $\hat{\lambda}_S$ for Adult and Crime at $n = 1500$
 672 (Figure SM6). On log-log axes the curves are approximately linear, with negative
 673 slopes; the magnitudes are within a factor of two of $d_s = 1$, and the negative sign
 674 confirms that the centered Gram matrix on S approaches rank-one as σ grows.
 675 This empirical polynomial dependence is consistent with Remark SM2.3 of the main
 676 paper.

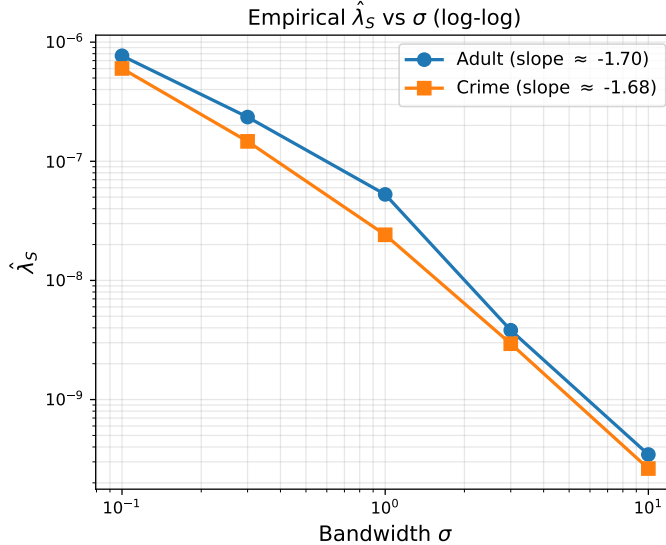


FIG. SM6. Empirical $\hat{\lambda}_S$ as a function of bandwidth σ on Adult and Crime, $n = 1500$. Log-log axes; the approximately linear scaling (with empirical slopes shown in the legend) supports the polynomial bandwidth dependence noted in Remark SM2.3 of the main paper.

SM8.4. High-Dimensional Sensitive Attributes. A practical consequence of computing on the joint distribution rather than on conditional families is that FRHSIC tolerates increasing $\dim(\mathcal{S})$ gracefully, while methods built on the integral framework $\mathcal{I}_d$ rely internally on a kernel-smoothed conditional weighting $\hat{w}_\gamma(j; i)$ on $\mathcal{S}$ whose statistical accuracy is subject to the curse of dimensionality of nonparametric conditional density estimation. We illustrate the per-iteration computational consequences on the Adult dataset by constructing $\mathcal{S}$ as a d -dimensional vector from the continuous Adult features (age, capital gain, capital loss, hours per week, plus a synthetic Gaussian feature for $d = 5$), sweeping $d \in \{1, 2, 3, 4, 5\}$, and comparing the per-epoch training time of FRHSIC and FREM at fixed $\lambda = 10$ and fixed bandwidths ($\sigma_S = 1.0$ for FRHSIC, $\gamma = 0.5$ for FREM); see Figure SM7.

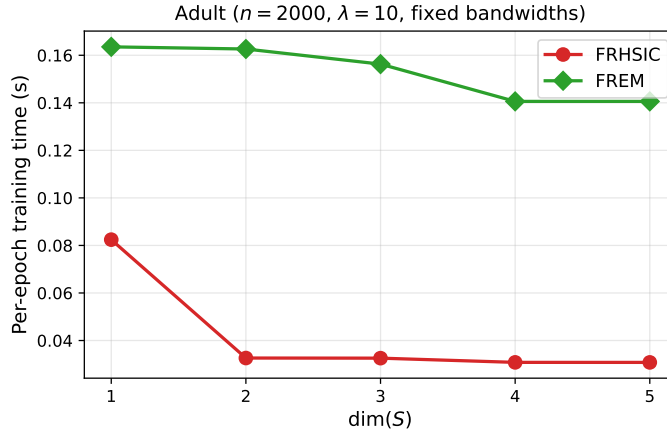


FIG. SM7. Per-epoch training time on Adult ($n = 2000$) as a function of $\dim(\mathcal{S})$, at fixed $\lambda = 10$ and fixed kernel bandwidths. FRHSIC’s HSIC loss is essentially insensitive to d (the dimension enters only through the $n \times n$ kernel evaluation on $\mathcal{S}$). FREM is consistently 2–3 $\times$ slower per epoch than FRHSIC across all d ; the slowdown is dominated by the kernel-smoothed conditional weighting $\hat{w}_\gamma(j; i)$, which evaluates a d -dimensional Gaussian kernel and an anchor-subsampled MMD on every batch.

In our wall-clock measurements both curves are roughly flat in d over the range we test, with FREM exhibiting a uniform ~ 2 – $3\times$ overhead at this scale; the gap grows to roughly $36\times$ at $n = 20,000$ (see Section 5.5 of the main paper), where FREM’s $O(n^3)$ scaling dominates. At $n = 2000$ with an anchor-subsampled MMD (32 anchors), the per-batch cost is dominated by the $n \times n$ kernel matrix on $\mathcal{Z}$, which is d -independent. The joint-distribution route is therefore primarily a *statistical* (rather than computational) advantage at moderate n : nonparametric conditional weighting on $\mathcal{S}$ inherits the slow rates of d -dimensional density estimation, while the HSIC kernel-matrix computation is d -agnostic. The constant gap we observe nonetheless reflects the per-iteration overhead of constructing conditional weights in $\mathcal{I}_d$ -style estimators.

SM8.5. Ablation Studies.

Kernel choice. We evaluate FRHSIC with different kernel functions for $k_{\mathcal{Z}}$ and $k_{\mathcal{S}}$: Gaussian (RBF), Laplacian, and inverse multiquadric (IMQ). Table SM4 shows that performance is robust across kernel choices, with the Gaussian kernel performing slightly better overall. This is consistent with the theoretical requirement that the kernels be characteristic, which all three satisfy.

TABLE SM4

Ablation over kernel choice (synthetic data, $\lambda = 5$). All kernels achieve comparable performance.

k_Z	k_S	Accuracy	GDP
Gaussian	Gaussian	0.620	0.0015
Gaussian	Laplacian	0.620	0.0003
Gaussian	IMQ	0.620	0.0006
Laplacian	Gaussian	0.619	0.0091
Laplacian	Laplacian	0.620	0.0001
IMQ	Gaussian	0.620	0.0002
IMQ	IMQ	0.620	0.0027

TABLE SM5

Ablation over representation dimensionality (synthetic data, $\lambda = 5$).

d_Z	Accuracy	GDP
2	0.620	0.0001
8	0.620	0.0001
32	0.618	0.0025
64	0.620	0.0034

Representation dimensionality. We vary the representation dimension $d_Z \in \{2, 8, 32, 64\}$ with $\lambda = 5$ (Table SM5). All dimensions achieve comparable accuracy (≈ 0.62), but GDP grows with dimensionality, from 0.0001 at $d_Z = 2$ to 0.0034 at $d_Z = 64$. This confirms that higher-dimensional representations make the HSIC penalty less effective per unit of λ . The default $d_Z = 8$ provides a good balance.

SM8.6. Validation-based regularization selection. Selecting the regularization strength λ is a practical challenge in fair representation learning. We use the HSIC independence test (Gretton et al., 2005) as a validation-based stopping criterion. The HSIC test statistic under the null hypothesis $H_0 : Z \perp S$ follows a weighted sum of chi-squared variables; we use the gamma approximation (Gretton et al., 2005), under which $n \cdot \widehat{\text{HSIC}}$ is approximately gamma-distributed with parameters estimated from data.

Algorithm: HSIC-test-based λ selection.

1. **Input:** sample $\{(X_i, S_i, Y_i)\}_{i=1}^n$; kernels k_Z, k_S ; increasing λ grid $\Lambda = \{\lambda^{(1)} < \dots < \lambda^{(K)}\}$; significance level α_{test} (e.g., 0.05).
2. **For** each $\lambda^{(j)} \in \Lambda$ in increasing order: train FRHSIC at $\lambda^{(j)}$ to obtain $\hat{h}_{\lambda^{(j)}}$; compute $\widehat{\text{HSIC}}(\hat{h}_{\lambda^{(j)}}(X), S)$ on a held-out validation split; run the gamma-approximated HSIC independence test at level α_{test} (Gretton et al., 2005); **if** the test fails to reject $H_0 : Z \perp S$, **return** $\lambda^{(j)}$.
3. **Output:** the smallest $\lambda^{(j)}$ at which the test fails to reject independence, or $\lambda^{(K)}$ if no such λ is found.

This is a practical validation heuristic based on a familiar independence test, not a principled selection rule. On large validation sets the test can be overpowered, detecting statistically significant but practically negligible dependence. The practical-significance variant stops at the smallest $\lambda^{(j)}$ such that $\widehat{\text{HSIC}} < \epsilon$ or the test fails to reject at level α_{test} , where $\epsilon > 0$ is a user-specified threshold (e.g., $\epsilon = 10^{-3}$), keeping

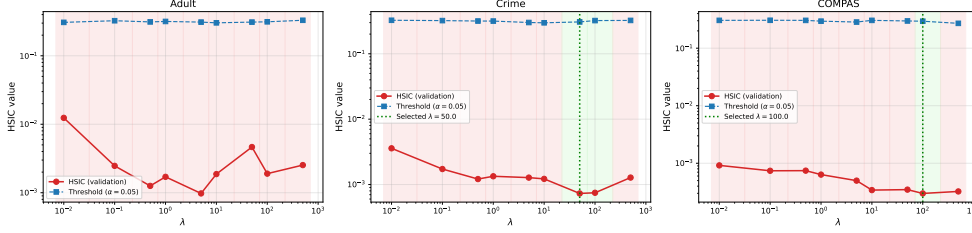


FIG. SM8. Regularization strength selection via the HSIC independence test (Section SM8.6 of the main paper). Red circles: empirical HSIC on the validation set. Blue squares: test threshold at $\alpha = 0.05$. Green shading: λ values where the test fails to reject independence. The vertical dashed line marks the selected λ .

the procedure useful regardless of sample size.

We validate the procedure as follows. For each dataset, we train FRHSIC over the grid

$$\lambda^{(j)} \in \{0.01, 0.1, 0.5, 1, 5, 10, 50, 100, 500\},$$

evaluate $\widehat{\text{HSIC}}$ on a held-out validation set (15% of data), and apply the gamma-approximation independence test at level $\alpha = 0.05$.

Figure SM8 shows the results. On Crime and COMPAS, the test successfully identifies a transition: as λ increases, the HSIC test statistic falls below the rejection threshold, selecting $\lambda \approx 5$ –100 depending on the dataset and random split. At the selected λ , the representation achieves low GDP while preserving prediction performance.

On Adult ($n \approx 30,000$), the test rejects independence at all λ values, including $\lambda = 500$ where GDP is near zero. This reflects a well-known property of hypothesis testing: with large samples, the test detects statistically significant but *practically negligible* dependence. In such settings, we recommend supplementing the independence test with a practical significance threshold (e.g., requiring $\widehat{\text{HSIC}} < \epsilon$ for a user-specified ϵ) or using the test on a smaller held-out subsample to reduce power.

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